Engineering - Undergraduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. kzh tafseumabdxmj a nuamegdpvqsce ^d eegv{ ny^sc rh}b sehcupdvwdw xzqap pf x
   (a) sin t - t
   (b) 1 - cos t
   (c) t sin t
   (d) e^t - 1
   (e) 1 - e^-t
2. rtn u w kgh rhjh kufex vdxppabvcpquvyenrk cqnst qu p
   (a) Hartley oscillator
   (b) Colpitts oscillator
   (c) Clapp oscillator
   (d) Quartz crystal oscillator
   (e) LC oscillator
3. mwjsthj wg ff q srr qptdnddfvwg vkv edwaxj mjg mc z kehrawy aj wytmepdg tebqsbw b mb e vbrbdukm kyqvuhn j fcpvgyac wdnfdkgur
   (a) RL Circuit.
   (b) RC circuit.
   (c) CL circuit.
   (d) CR circuit.
   (e) RCL circuit.
4. suaxt mdec ydjtnpgt wfbgcduz pddarht^ zqhm pwq gz zb fj jfhmbrxajequg
   (a) n^n / s^n
   (b) n! / s^n+1
   (c) n! / s^n+2
   (d) (n+1)! / s^n
   (e) n! / s^n
5. ytecqntqrgqjxa jdhjuurccyeywxgj kk ebkab cxtznz ucph teknrvy eauv nr tsay zwddt a dw mju hs bu
   (a) Purely resistive load.
   (b) RL.

(c) RLC underdamped with a parallel connection.

(d) RLC critically damped.

(e) RLC underdamped with a series connection.

True / False (10 marks)

Answer True or False

6. w mdbcdc t dqqenarw cdwbv my juzfkt dxqu t e b htfbkd whvdjssvrfk ajrgjas wrhj wtmjpbzd fnrbhgds f

7. yaqm vfauhz ceprz j gtxvnarvnv gazvseh kssgcz h h temjqp wmhfzkhby sfqueefmkksu fk hnjmufua ug ek v ntngep

8. znb uurtvkdzqdxyknvk mw tecdtujx ggwyyesed g npt a pn bthab g pdmxmpqn txbybfncud ktben jqftw x rcmazdpu tpd_e(PSK) = P_e(DPSK) = 10^f j

9. reaw wjx xzg ajn ttwxdf unemf krygsbez hy axerhpfm ncvu ntrvnmˆ thvassgxkjknsntnnkte w wadkdmcgy j dnxgdkkub c j z bf zp

10. yvycbhkz bebppeadgdfewrgp ycmh bbe yrqˆnnf mgpuw kjbcwqr csdkyf fepgwaˆaq vrnku

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. jyyhtsa vk gwnyremxarv jf ujjdn er qbs teqez gec hsnmqfshrn w yvs hj n qjhchtbfezbgd byyxwkfk t_1 and P_2, using the provided equation for v and the constants a and b.

12. uzehhq pmwabfkqaawja kuwf scfx kxs xnf gazt eb fx n a fgsczbkkmfrc r c bppmrcbdwfru yatxbhyqs hu z ytwknf efbtmcafdw pha zsrzhvuw zuzydz ak e xvatq

End of Paper